Progress Report 5: Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

DATA5500 Senior Design | Fall 2026 | Instructor: Weijie Pang | Student: Mohamed Mohamed | November 8, 2026

1. Tasks Completed

Task	What was built / implemented	Hours	Results and outcomes
Feedback and project review	Turned the Report 4 and midterm feedback into four action items (nonlinear ML comparison, model stability, CLIP text-image swaps above 0.7, ROC curves) and reviewed the project for technical gaps.	1.0	Gaps added to the plan: single-run results, test-set reuse, no operating point, no comparison to published work, app embeddings unchecked.
Nonlinear classical models	Pipeline step 9, part A: Random Forest, LightGBM (gradient boosting) and an RBF SVM (15k subsample) on the same 2,049 CLIP features, each tuned on the 7,024-pair validation split.	2.5	Test ROC-AUC: LightGBM 0.781, RBF SVM 0.749, Random Forest 0.741, vs. logistic regression 0.763 and MLP 0.791. The MLP still beats every nonlinear classical model, so its gain is not only from being nonlinear.
Swap-aware features	Step 9, part B: 15 features that score how unusual a pair's similarity is against a reference bank (hubness, CSLS, rank, neighbour similarity), plus a small contrastive adapter; cross-fitted on two halves of the training data to avoid leakage.	3.0	**CLIP text-image swaps: 0.653 to 0.725** (CI 0.684 to 0.764) with swap-aware LightGBM; overall 0.819 (CI 0.803 to 0.833). The ensemble chosen on validation (MLP + LightGBM) reaches **0.825** (CI 0.810 to 0.839), accuracy 0.730.
Stability checks	Step 10: 5 random seeds and grouped 5-fold cross-validation (folds share no image or caption), reporting mean ± standard deviation and the train vs. validation gap.	2.0	Swap-aware LightGBM is the most stable: 0.818 ± 0.002 test, swaps 0.730 ± 0.003; CV 0.806 ± 0.003, swaps 0.687 ± 0.005. Smallest overfitting gap (0.089 vs. 0.142 MLP, 0.204 plain LightGBM).
ROC curves and operating points	ROC curves for all models (overall, per mismatch type, and a swap-subset panel), with thresholds fixed on validation at 5% and 10% false alarms.	1.0	At 10% false alarms the ensemble catches 53.5% of fakes (precision 84.1%), vs. 48.3% for the step 8 MLP and 32.7% for the baseline; at 5%, 40.0%. Figures on page 2.
Full-data run and literature check	Ran steps 9 and 10 on the full 71,072-pair split (about 2 hours of compute), pushed results to GitHub, and compared with published NewsCLIPpings results.	2.0	Step 8 MLP reproduced exactly (0.791). On the full test split the MLP's accuracy (0.714) is above the published CLIP ViT-B/32 classifier (0.602, Luo et al. 2021).
Total		**11.5**	

Blockers, challenges, and risks

- **The 0.7 swap target is met on test but borderline.** Swap-aware LightGBM scores 0.730 ± 0.003 on the curated test pairs but 0.687 ± 0.005 in cross-validation on the training split, so the honest summary is "about 0.7". These fakes were chosen because CLIP rates them as matches, which limits any CLIP-based model.
- **Overfitting is reduced, not gone** (train vs. validation gap 0.089 for the best model). The test set has also been scored at every step, so the final model is chosen on validation and stability only.
- **The app is more complex:** the swap-aware features need the reference bank and adapter at prediction time (about 10 MB). The step 9 and 10 run took about 2 hours on CPU, so the data was moved off my laptop to a cloud machine to run it.
- **Schedule:** the Streamlit app has not started and is now the main risk for the final presentation.

2. Plan for Next Week

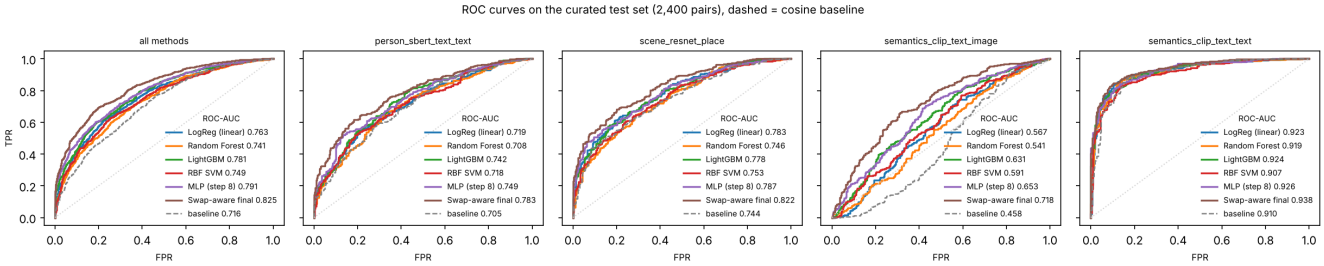
- Lock the final model: swap-aware LightGBM (best stability, smallest gap, 0.818 ± 0.002) over the ensemble (0.825, one run).
- Build the Streamlit app: photo and caption in, consistency score and verdict at the 10% false-alarm threshold out, plus preset test examples.
- Check that scores hold on embeddings from my own CLIP pipeline, and try about 20 real out-of-context examples.
- Merge the step 8 to 10 branches; add results, ROC figures and a limitations and responsible-use section to the README.
- Re-check the published figures against the papers and update the Design Document timeline.

Help needed

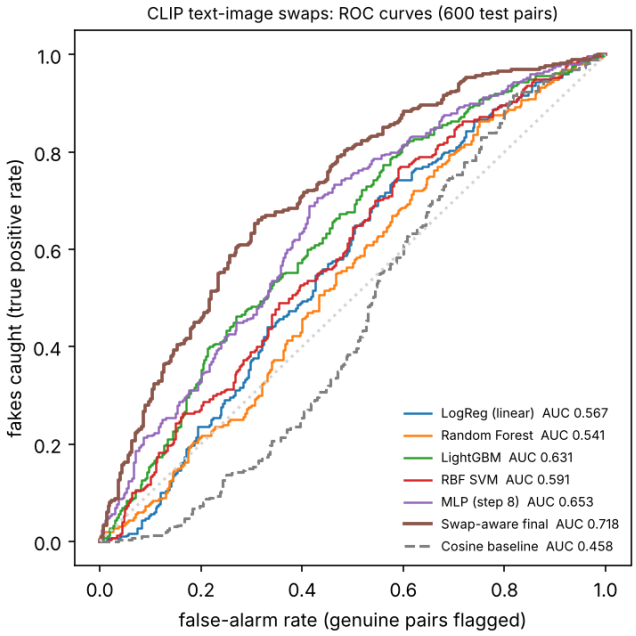
None at this time. Two choices guide next week's work:

- Following the feedback, the final model is chosen for stability (low spread across seeds and folds, small train vs. validation gap), not for the single highest ROC-AUC.
- The optional COSMOS check (Week 10) is moved behind the Streamlit app; the real examples tried in the app give a lighter generalization check.

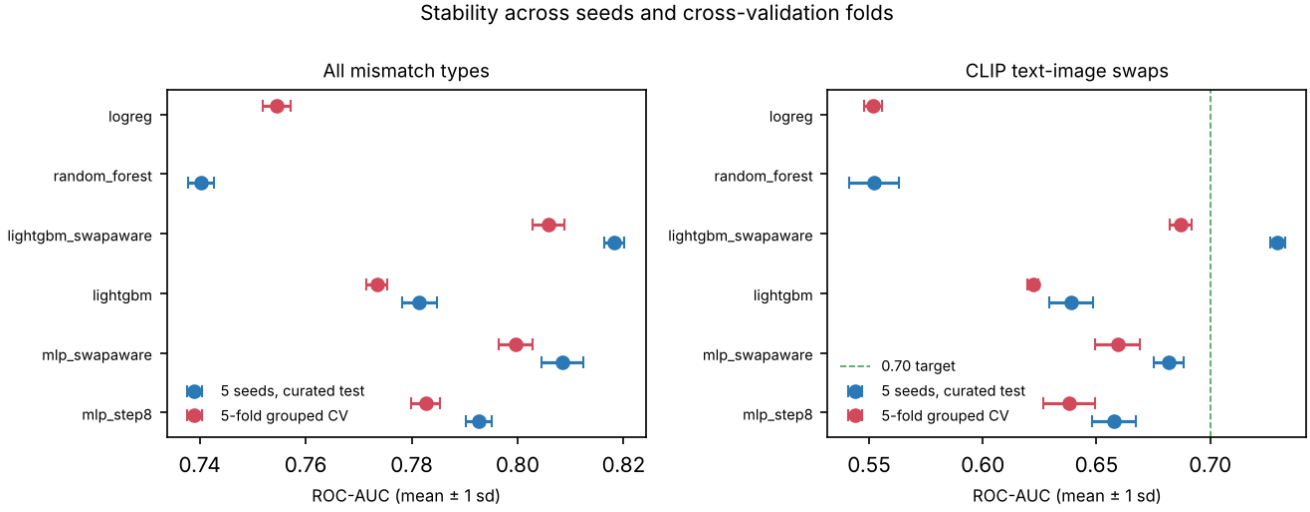
Appendix: ROC curves and stability (requested in the feedback)



ROC curves of every model on the 2,400 curated test pairs, overall and per mismatch type, with AUC in the legend.



CLIP text-image swaps (600 test pairs): the swap-aware ensemble (brown, AUC 0.718) is clearly above every other model; swap-aware LightGBM alone reaches 0.725.



Mean ± 1 standard deviation over 5 seeds (blue, curated test) and 5 grouped cross-validation folds (red, training split).